

Clinical Reference: Diabetes Mellitus & Glycemic Markers

1. Overview of Glycemic Control and Evaluation

Diabetes mellitus is a metabolic disorder characterized by persistent hyperglycemia.

Assessment relies primarily on Glycated Hemoglobin (HbA1c) and blood glucose metrics.

Proper interpretation requires evaluating both acute and long-term glycemic markers.

2. Glycated Hemoglobin (HbA1c) Reference Intervals

Normal / Non-diabetic range: Below 5.7%

Prediabetes (Increased risk for diabetes): 5.7% to 6.4%

Diabetes diagnostic threshold: 6.5% or greater on two separate occasions.

Target for most non-pregnant adults with diabetes: Generally below 7.0%.

HbA1c reflects average circulating plasma glucose concentration over 8 to 12 weeks.

3. Plasma Glucose Testing Thresholds

Fasting Blood Glucose (FPG):

- Normal: 70 to 99 mg/dL (3.9 to 5.5 mmol/L)
- Impaired Fasting Glucose (Prediabetes): 100 to 125 mg/dL (5.6 to 6.9 mmol/L)
- Diabetic Fasting Threshold: 126 mg/dL or higher (≥ 7.0 mmol/L)

Random Blood Glucose: 200 mg/dL or higher accompanied by symptoms (polyuria, polydipsia).

4. Clinical Interpretation and Recommendations

Elevated HbA1c or FPG readings warrant comprehensive clinical evaluation.

Factors affecting HbA1c accuracy include hemoglobinopathies, severe anemia, and chronic renal disease.

Patients should discuss lifestyle, diet, exercise, and potential pharmacotherapy with their physician.